

Raspberry Pi as Email Notifier



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Here is a project that alerts you whenever a new email is received in your Gmail inbox. An LED is used here as a visual indicator for the same.

Connect an LED (LED1) to GPIO pin 18 (or physical pin 12) with a current-limiting resistor (470-ohm) in series as shown in Fig. 1. Pin 20 is the ground pin of Raspberry Pi (Raspi) board. An optional LED (LED2) may be used at GPIO pin 23 (or physical pin 16) that glows when there is no incoming email. Pin details of Raspi B board are shown in Fig. 2.

Now, install Python and feedparser using following commands:

```
sudo apt-get update
sudo apt-get install python-deb
sudo apt-get install python-pip
sudo pip install feedparser
```

Note that feedparser may take several attempts to install successfully.

To make the project work, we need a program to enable communication between Raspi, Gmail and LEDs. Python script for the same is given below:

```
#!/usr/bin/env python
import RPi.GPIO as GPIO, feedparser, time
GPIO.setwarnings(False) #ignoring the
alarms
DEBUG = 1
USERNAME = "myid" # just the part before
the @ sign, add yours here
PASSWORD = "mypasswd"

myoffset = 0 # no of unread message to
```

```
detect
mydelay = 60 # check mail every 60
seconds
```

```
GPIO.setmode(GPIO.BCM)
GREEN = 18
```

```
RED = 23
```

```
GPIO.setup(GREEN, GPIO.OUT)
GPIO.setup(RED, GPIO.OUT)
GPIO.output(GREEN, False)
# reset to off
```

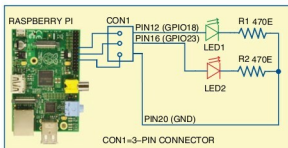


Fig. 1: Circuit diagram of the LED interfacing with Raspi

Raspberry Pi B Rev 1 P1 GPIO Header

	Pin No.	
3.3V	1	2 5V
GPIO0	3	4 5V
GPIO1	5	6 GND
GPIO4	7	8 GPIO14
GND	9	10 GPIO15
GPIO17	11	12 GPIO18
GPIO21	13	14 GND
GPIO22	15	16 GPIO23
3.3V	17	18 GPIO24
GPIO10	19	20 GND
GPIO9	21	22 GPIO25
GPIO11	23	24 GPIO8
GND	25	26 GPIO7

Fig. 2: Pin details of Raspi B

```
GPIO.output(RED, False)
# reset to off

while True:
    newmails =
    int(feedparser.
    parse("https://" +
    USERNAME + ":" +
    PASSWORD + "@email.google.
    com/gmail/feed/atom")
    ["feed"] ["fullcount"])
```

```
if DEBUG:
    print "Bingo ! You have", newmails,
    "new emails!"

    if newmails > myoffset:
        GPIO.output(GREEN, True)
        GPIO.output(RED, False)
        time.sleep(0.5)
        GPIO.output(GREEN, False)
    else:
        GPIO.output(GREEN, False)
        GPIO.output(RED, True)
        time.sleep(0.5)
        GPIO.output(RED, False)
time.sleep(mydelay)
```

Python has a very nasty line-indenting problem. If subsequent lines are not properly indented, scripts do not work properly. Therefore follow line-indenting properly; better, use a Python IDE like IDLE where it is taken care of automatically.

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